The Hippocampus

Part 1: Anatomy and Overview

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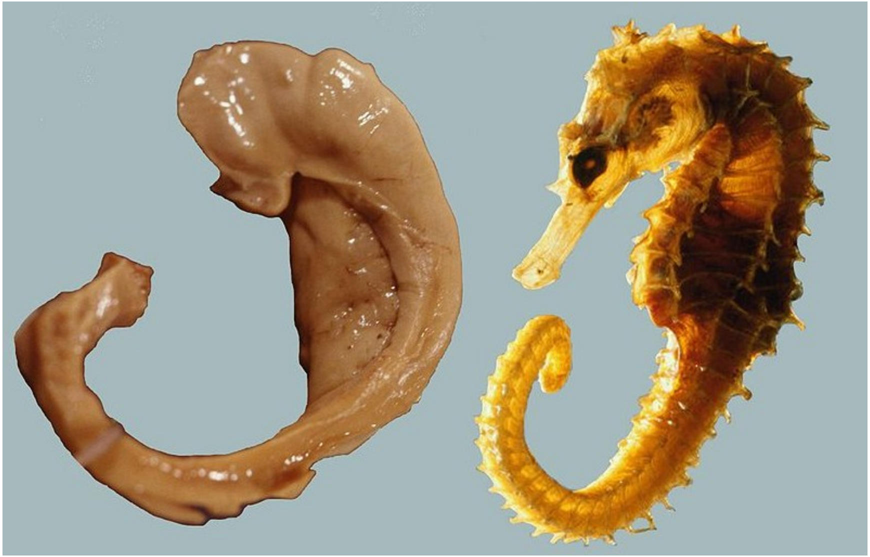
mrule

Learning outcomes:

- Be able to describe basic anaotmical location and features of the hippocampus
- Be able to name two key cognitive functions of the hippocampus
- Become familiar with hippocampal regions we will study in the context of memory an navigation

Hippocampus = (horse + sea monster)

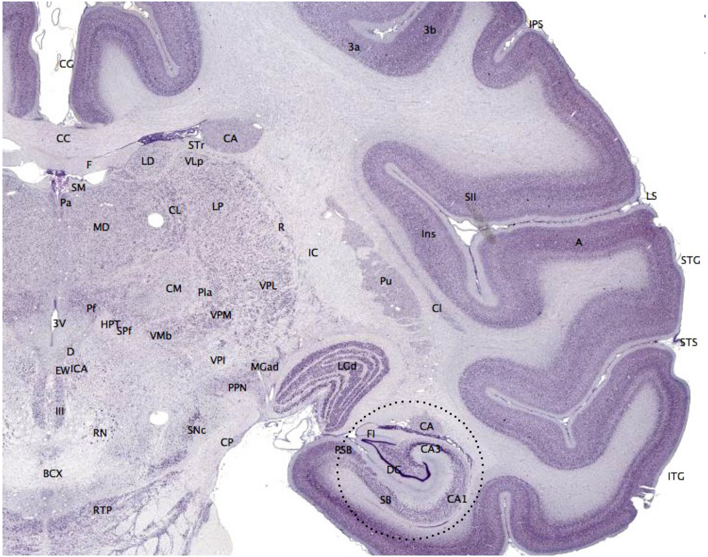
CA = Cornu Ammonis



[*L Seress' dissection of hippocampus and fornix, and a seahorse*]

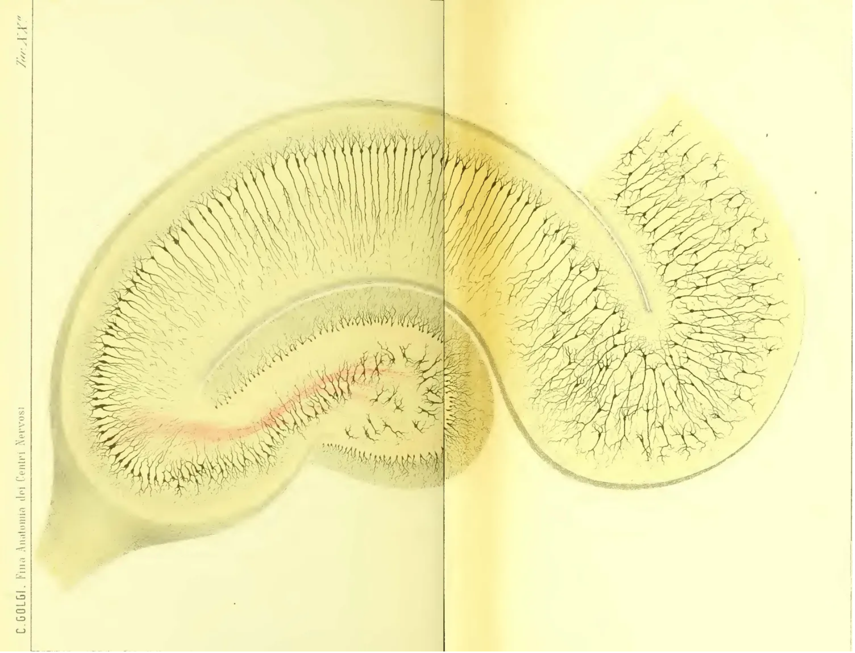
https://mouse.brain-map.org

Circle the hippocampus



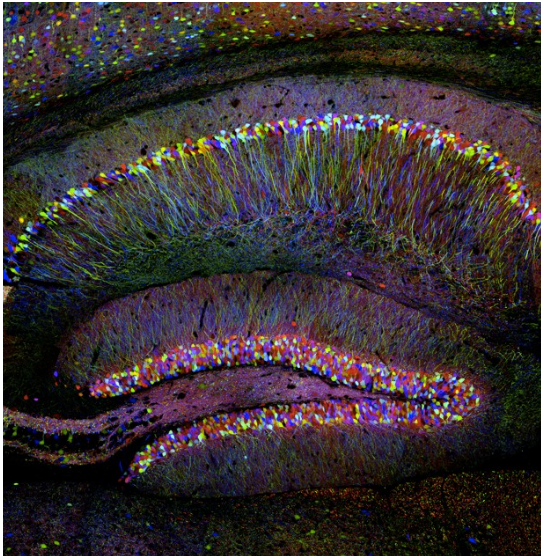
brainmaps.org

Circle the hippocampus



Label the parts and say what they do

Hippocampus, Camillo Golgi

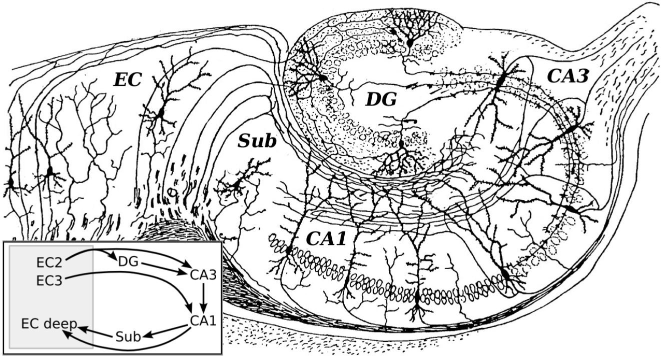


“Brainbow”, Livet & al. (2007) Nature

Hippocampus

Archicortex, deep inside temporal lobe

You need it to **form new memories** and **navigate the world**



Modified from Cajal (1911)

Entorhinal Cortex (EC)

Dentate Gyrus (DG)

***Cornu Ammonis* field 3 (CA3)**

***Cornu Ammonis* field 1 (CA3)**

Subiculum (Sub)

We will discuss these more in context of ***memory*** and ***navigation***

The tri-synaptic loop

Entorhinal cortex (EC): Gateway

Trisynaptic loop

- **Dentate Gyrus (DG)** $\longrightarrow$ **CA3** $\longrightarrow$ **CA1** $\longrightarrow$ **Subiculum**

CA3 the ***only*** subregion with ***substantial recurrent excitatory connectivity***

- recurrent **attractor network**
- pattern completion

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